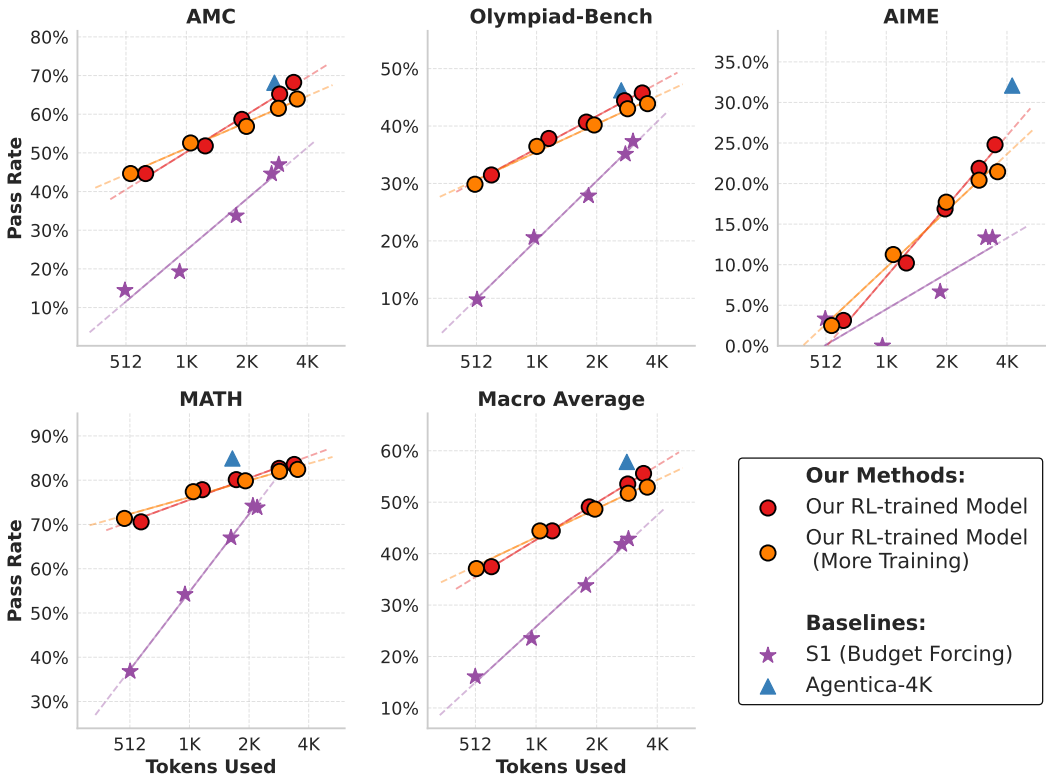

Figure 8: Performance vs Length for LCPO with more training

A.2 LCPO follows length constraints with high precision.

In Section 5, we show that LCPO follows length constraints with high precision, although with larger errors in OOD datasets. In this section, we particularly analyze the generation length at which error occurs. Figure 9 shows error as a function of generated token length for 3 different OOD datasets.

Further, we also report both soft- and hard-violation rates in Figure 12 and Figure 13. Notably, while soft-violation rates are very low ($\leq 3\%$), hard-violation rates are relatively higher (9% on average across different token budgets). The violation rates are particularly higher at lower token budgets, as the model prioritizes performance over length control, and generating correct solutions with shorter CoTs is relatively more difficult.

Model	AIME2024	AIME2025	Delta
Qwen-Math-1.5	12.9	7.7	-5.2
GPT-4o	6.7	10.0	+3.3
LLama-3.3-70B	27.3	3.8	-23.5
DeepSeek-R1-1.5B	29.2	21.7	-7.5
DeepScaleR-24K	**40.2**	27.3	-12.9
DeepScaleR-4K	32.1	25.4	-6.7
L1-Exact *(ours)*	24.8	22.5	-2.3
L1-Max *(ours)*	27.3	21.5	-5.8
L1-Exact *(1024 tokens)*	10.2	12.3	+2.1
L1-Max *(1024 tokens)*	16.3	11.9	-4.4

Table 3: Comparison of model performance on AIME2024 and AIME2025 benchmarks.

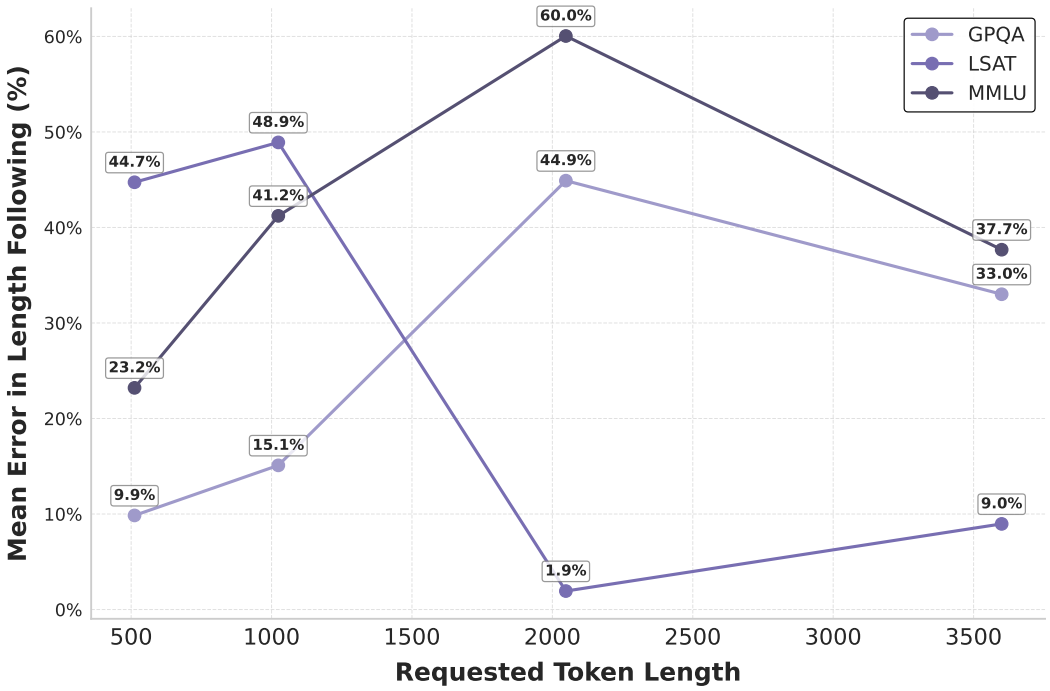

Figure 9: Length error vs token length for OOD datasets. Datasets like MMLU show higher error for longer chains, likely because longer chains are not particularly useful for these datasets.

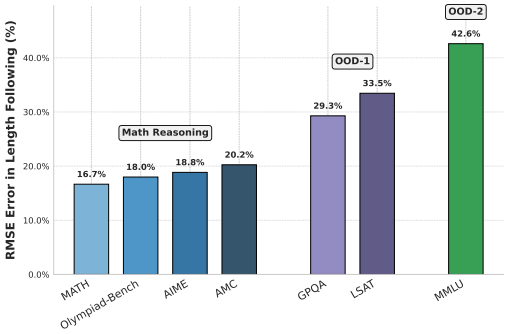

Figure 10: **RMSE in Length of generated chains** across different datasets. LCPO exhibits low RMSE for math reasoning datasets, maintaining high precision in length control. OOD datasets exhibit higher RMSE, particularly MMLU, where longer chains are less necessary.

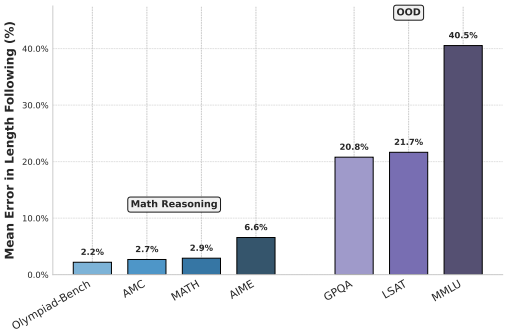

Figure 11: **Mean Error in Length of generated chains** across different datasets. L1 maintains around 3% average deviation on math reasoning datasets and higher but acceptable deviation on OOD tasks.

Memorization or Generalization? Given the surprisingly strong performance of LCPO, particularly at lower token budgets, a natural question arises regarding whether the model has memorized solutions, suggesting potential train-test leakage. To address this concern, we specifically evaluate on the AIME2025 dataset, an exam released after the training of all evaluated models. Results in Table 3 demonstrate that the performance drop for LCPO is substantially smaller ($\approx 2\%$) compared to other models such as DeepScaleR-24K (-12.9%), DeepSeek-R1-1.5B (-7.5%), and even Llama-3.3-70B (-23.5%). These results provide strong evidence that despite using shorter CoT lengths, our model demonstrates generalization rather than relying solely on memorization.

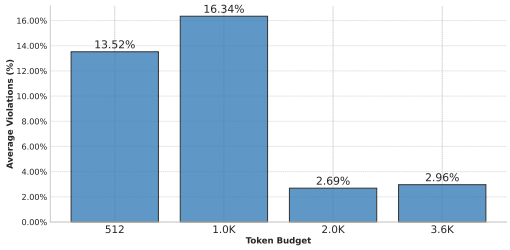

Figure 12: **Hard Budget Violation Rates** across different datasets. L1 maintains relatively higher, but acceptable violation rates of 9% on average across different token budgets.

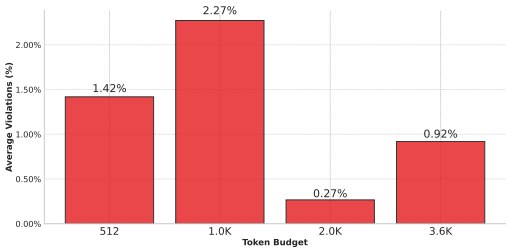

Figure 13: **Soft Budget Violation Rates** across different datasets. L1 maintains less than 3% budget violation rate on across different token budgets.

A.3 Aha Moment in L1's training

Figure 14 shows the training logs of L1-Exact. During the first 300 RL steps, we observe that the model prioritizes improving its average solve rate while the minimum response length continues to decrease. This suggests that the model initially focuses on correctly generating solutions rather than strictly adhering to length constraints. However, at approximately 300 RL steps, we observe a significant phase transition in the training dynamics. Specifically, the reward and token adherence score (reward $-$ solve_mean) begin to increase rapidly, while the minimum response length exhibits a sharp drop, eventually converging at around 100 tokens (the lower bound of n_{gold}). This distinct transition point represents when the model effectively learns the concept of token adherence and subsequently continues to refine this capability throughout further training.

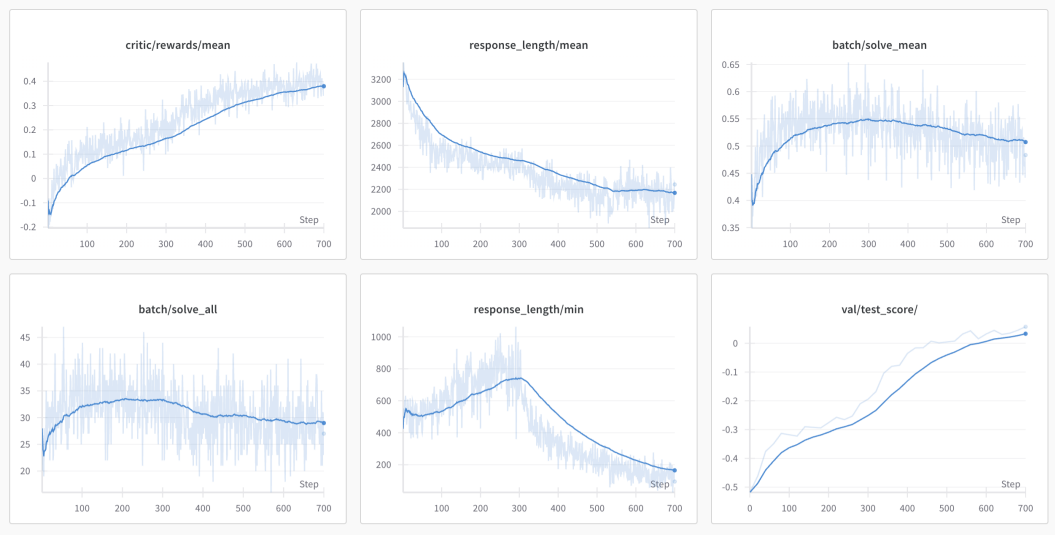

Figure 14: Training Logs for LCPO-Exact. We notice an Aha Moment, where the model starts adhering to token constraint after 300 RL steps, as visible in peaking solve rate, and sharp drop in minimum token length.

A.4 Does L1 generalize to varying sampling parameters

We evaluate L1-Exact's robustness to different sampling parameters by varying the temperature during inference. Table 4 presents results across temperature values ranging from 0.0 to 1.0. We observe that the model maintains consistent performance in terms of both solve rate and length control accuracy. Specifically, the mean deviation from target length